

ECON 0100 | Exercise B5 | Market Changes

I Solutions

Answers and work are shown in the red blocks.

Pumpkin pasties again, with the same demand and supply curves as Exercise B3:

$$P = 12 - \frac{Q_d}{2} \qquad P = 2 + \frac{Q_s}{2}$$

Prices are in galleons and quantity is in pasties. The equilibrium you found is 10 pasties at 7 galleons.

Q1 | Price Elasticity

- a) Use the midpoint method to find the elasticity of demand when the price changes from 8 to 10 galleons. _____
- b) Is demand elastic, unit elastic, or inelastic in a)?
- c) Use the midpoint method to find the elasticity of demand when the price changes from 2 to 4 galleons. _____
- d) Use the midpoint method to find the elasticity of supply when the price changes from 3 to 5 galleons. _____

a) -3 . b) Elastic. c) $-\frac{1}{3}$. d) 2.

The midpoint method divides each change by the average of its two endpoints, so the answer is the same whichever way the price moves:

$$\varepsilon = \frac{\% \Delta Q}{\% \Delta P} = \frac{(Q_2 - Q_1) / \frac{Q_1 + Q_2}{2}}{(P_2 - P_1) / \frac{P_1 + P_2}{2}}$$

a) Prices to quantities along demand: $P = 8$ gives $Q = 8$ and $P = 10$ gives $Q = 4$.

$$\varepsilon_D = \frac{-4/6}{2/9} = -\frac{2}{3} \cdot \frac{9}{2} = -3$$

b) $|-3| > 1$: a 1% rise in price cuts quantity demanded by 3%, so demand is elastic between these prices.

c) $P = 2$ gives $Q = 20$ and $P = 4$ gives $Q = 16$.

$$\varepsilon_D = \frac{-4/18}{2/3} = -\frac{2}{9} \cdot \frac{3}{2} = -\frac{1}{3}$$

Inelastic. The slope is the same everywhere on a straight demand curve, yet elasticity is high up near the intercept and low down near the quantity axis: the same 2-galleon, 4-pasty step is a big percentage of a small price and a small percentage of a large quantity.

d) Along supply, $P = 3$ gives $Q_s = 2$ and $P = 5$ gives $Q_s = 6$.

$$\varepsilon_S = \frac{4/4}{2/4} = 1 \cdot 2 = 2$$

Elastic, and positive, as the law of supply says.

Q2 | Supply & Demand Shifters

A subcommittee of the Ministry published a story in the Daily Profit establishing a link between the consumption of pumpkin pasties and accidental magical spell casting by wizards and witches in public areas.

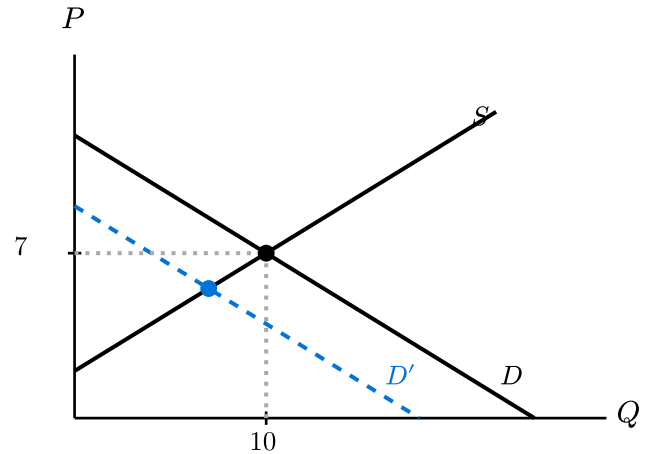
a) The demand curve: *shifted up, stayed the same, shifted down*

b) The supply curve: *shifted up, stayed the same, shifted down*

a) Shifted down. b) Stayed the same.

The study lowered buyers' willingness to pay for pumpkin pasties at every quantity, shifting the demand curve down. Nothing about the story changes what it costs to bake a pasty, so supply stays where it is.

Along the unchanged supply curve, the lower demand lowers both the price and the quantity: the new equilibrium sits down and to the left of the old one.



Q3 | Comparative Statics

Pumpkin growers suddenly discover a miracle fertilizer that significantly increases harvests. On it's own, how did this discovery impact the market?

a) The equilibrium price: *increased, stayed the same, decreased*

b) The equilibrium quantity: *increased, stayed the same, decreased*

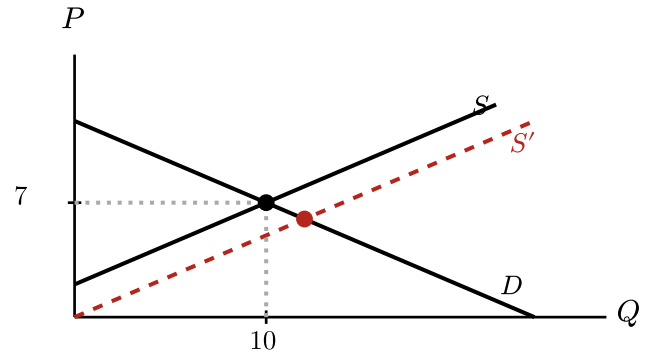
Without using numbers, how have these two changes together (the study and the fertilizer) impacted the pumpkin pasty market equilibrium?

c) Equilibrium price has: *increased, stayed the same, decreased, indeterminate*

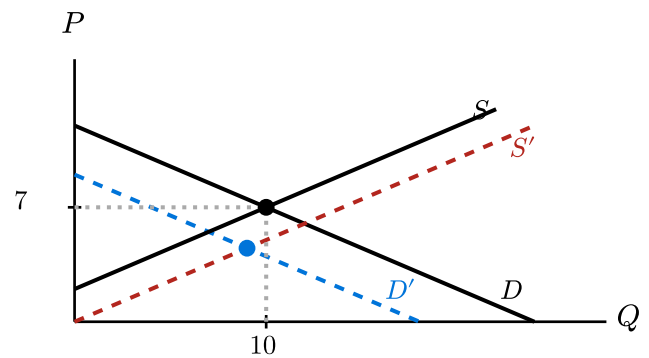
d) Equilibrium quantity has: *increased, stayed the same, decreased, indeterminate*

I a) Decreased. b) Increased. c) Decreased. d) Indeterminate.

a), b) Bigger harvests make pumpkins cheaper for bakers, lowering the marginal cost of every pasty: supply shifts down. Along the unchanged demand curve the price falls and the quantity rises.



c), d) Put the two shifts together. Demand down pushes the price down, and supply down pushes the price down: the price has decreased for certain. Demand down pushes the quantity down, but supply down pushes it up, so without numbers the quantity is indeterminate; it depends on which shift is bigger.



The graph draws one case: the demand shift happens to outweigh the fertilizer and the quantity ends slightly lower. A bigger harvest would have pushed it the other way, which is why the answer without numbers is indeterminate. Either way the price is lower.